PROJECT SUMMARY REPORT

Precision Cargo Delivery UAV — Autonomous Navigation & Object Detection

AeroKLE Aero Design Team · KLE Technological University, Hubballi

SAE INDIA AEROTHON
TECHNICAL DESIGN REPORT-2022

TEAM NO. 49

Figure 1 — Final UAV: X-Configuration Quadcopter (SolidWorks CAD Model) for autonomous cargo delivery

Competed	< 2 kg	200 g	9.87 min	1 km+	1780 g	AIR 9
SAE Aerothon 2022	Max Takeoff Weight	Payload Capacity	Endurance	Comms Range	Empty Weight	Among 50+ Teams

SKILLS DEMONSTRATED BY TEAM

CAD & Modelling	SolidWorks — full parametric assembly, CG calculation, moment of inertia analysis
Structural FEA	ANSYS Mechanical — static structural analysis on 7 components, topology optimisation, material selection
Control Systems	MATLAB Simulink — quadcopter dynamics model, PID tuning (4 axes), Fuzzy-PID hybrid controller
Autonomous Systems	ROS/Gazebo simulation, MAVLink/MAVros, DroneKit (Python), Pixhawk integration
Computer Vision	YOLO v4-tiny implementation on Raspberry Pi 4 for real-time single-class object detection
Manufacturing	3D printing (PLA Composite/ABS components), carbon fibre tube cutting, hardware integration
Python Programming	DroneKit autonomous mission scripting, state machine implementation, crash prediction ML model

Note: This document serves as a summary for portfolio purposes. Complete technical report contains proprietary information & is protected*

1. COMPETITION OVERVIEW

The SAE India Aerothon 2022 is a national-level UAV engineering competition where student teams design, build, and fly a stable multirotor UAV capable of autonomous cargo delivery. Teams are evaluated on their Technical Design Report, oral presentation, and actual flight performance — including autonomous take-off, GPS navigation, computer-vision target identification, precision payload drop, and return-to-home.

1.1 Problem Statement & Competition Requirements

UAV Type	Multirotor (Micro UAS category)
Max takeoff weight	< 2,000 g
Payload capacity	200 g — enclosed in 10 × 5 × 5 cm bay
Propulsion	Electric — brushless DC motors
Comm. frequency	Data telemetry: 2.4 GHz | Video telemetry: 2.4 GHz or 5.8 GHz
Comm. range	≥ 1 km
Flight modes	Both manual and fully autonomous
Autonomy requirements	Take-off → GPS navigation → target detection → payload drop → land at start point

1.2 Mission Profile

The UAV must autonomously take off, cruise at 30 m altitude, search a 30 × 20 m target area, detect and identify the target using onboard computer vision, descend to 20 m, drop the payload precisely on target, then navigate back to the launch point and land safely.

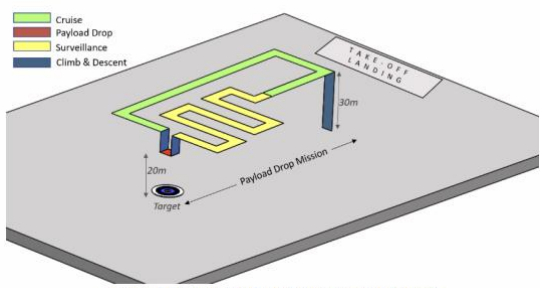

Figure 1 Mission Profile of the designed aircraft

1.3 My Contribution

I was a core designer on the team, contributing primarily to structural analysis (ANSYS FEA on all major components), CAD modelling in SolidWorks, material selection, and the payload drop mechanism design and analysis. The project was a valuable introduction to UAV systems integration, multi-physics analysis, and autonomous system architecture.

2. CONCEPTUAL DESIGN & CONFIGURATION SELECTION

A comprehensive literature review was conducted across tricopter, quadcopter, pentacopter, hexacopter, and octacopter configurations. The Quad-X symmetric configuration was selected after a weighted trade-off covering stability, power consumption, structural strength, manufacturability, and manoeuvrability — scoring 77/80, the highest of all options.

Table 2 Comparison Chart for Quadcopter Configuration

Parameters	Total score /5	Symmetric				Asymmetric			
		Quad X		Quad +		Quad-Stretched		Quad v tail	
		Score	Total	Score	Total	Score	Total	score	Total
Stability	5	4	20	3	15	3	15	5	25
Power consumption	4	4	16	3	12	4	16	2	8
Structural Strength	4	5	20	4	16	5	20	3	12
Manufacturability	3	4	12	4	12	3	9	3	9
Manoeuvrability	3	3	9	4	12	4	12	2	6
Total			77		67		72		60

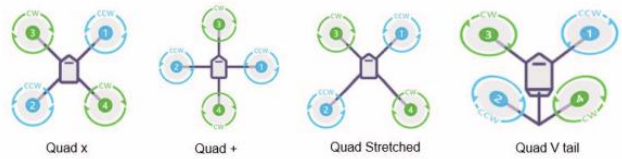

Figure 2 Types of Quad Configurations

2.1 Propulsion System Design & Endurance calculation

With a target MTOW of 2 kg, a T/W ratio of 2:1 was specified (industry standard for stable, manoeuvrable multirotors), requiring 4 kg total thrust — 1 kg per motor. Four motors were evaluated across thrust, weight, max current, and endurance.

Selected motor	SunnySky V2806 (400 KV) — best performance-per-watt ratio
Propeller	12 × 5 inch APC
Thrust per motor	1,100 g static | T/W per motor: 1.09
Total thrust	4,400 g (T/W ratio = 2.2 at MTOW 2 kg)
ESC	Hobbywing Xrotor Nano 20A 4-in-1 BLHeli DShot600 (hub-mounted)
Battery	4S LiPo, 3,700 mAh, 45C discharge rate (calculated min: 3,828 mAh)
Motor voltage	14.8 V
Endurance	9.87 min at full MTOW (target: ≥ 10 min)

$$Time = \frac{Capacity\ X\ Discharge}{AAD}$$

$$Time = \frac{3700.9}{AAD} \qquad AAD = \frac{AUWP}{V}$$

$$AAD = \frac{2kg\ X\ 170}{16.8} = 20.238$$

$$t = \frac{3.7Ah\ X\ 9}{20.238} = 9.872min$$

The 4-in-1 ESC was specifically chosen over four individual ESCs — mounting it in the central hub protects it from crash damage and prop strikes, while also allowing slimmer arm profiles to reduce aerodynamic drag.

3. STRUCTURAL DESIGN

3.1 Rotor Hub Architecture

A modular design was adopted over a unibody frame — individual carbon fibre arms are separately attached to the mid-structure. In the event of a crash, only the damaged arm needs replacement rather than the entire airframe, significantly reducing repair cost and turnaround time.

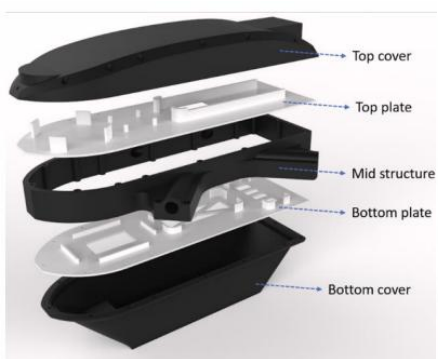

Figure 6 Exploded view of Hub and its components

Figure 9 Landing Gear mounting position

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3.3 Material Selection

Table 9: Comparison of material properties

Material	Tensile Strength (MPa)	Young's Modulus (GPa)	Flexural Strength (MPa)	Fracture Behavior	Density (g/cm³)
Carbon Nylon	46.4	4.66	110	Brittle	1.55
PLA	38.4	2.87	87	Brittle	1.24
ABS	19.6	1.46	77	Ductile	1.04
Carbon Fiber	989	245	310	Brittle	1.8

Table 10: Justification for selected materials

Part	Material	Justification
Avionic bay, motor mount, top shell, bottom shell and payload mechanism	PLA	Durable, rigid, easy to manufacture and cost effective
Payload plate	Birch ply	Strong, durable, and fatigue resistance.
Motor Arms, Landing gear	Carbon fiber	High strength, high impact resistance, lightweight and durability.

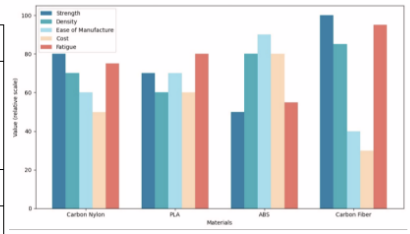


Figure 17: Material property comparison of carbon nylon, PLA, ABS and carbon fiber

4. FINITE ELEMENT ANALYSIS (FEA) — COMPONENT VALIDATION

All major structural components were individually analysed in ANSYS Mechanical under their respective loading conditions. Every component was validated to have FOS ≥ 1.5 in its final material assignment. Results directly drove the design optimisation decisions in Section 5.

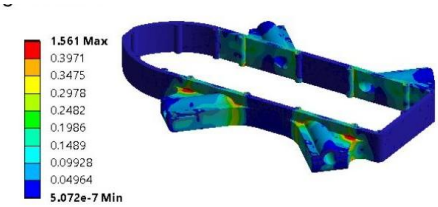


Figure 25 Stress distribution on mid structure

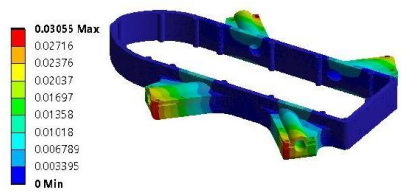


Figure 26 Deformation of mid structure

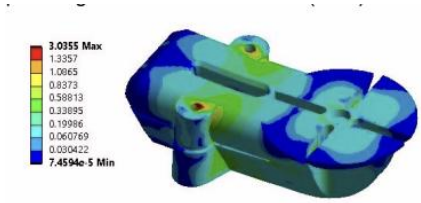


Figure 30 Stress distribution on Motor Mount

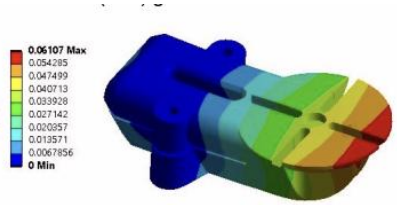


Figure 31 Deformation of Motor Mount

5. SUB-SYSTEM DESIGN — PAYLOAD MECHANISM

5.1 Payload Drop Mechanism

Four door mechanisms were evaluated — slider doors, double doors, double slider, and single hinged — based on actuation simplicity, manufacturing ease, weight, accuracy, and response time. A single servo-actuated hinged door was selected, mounted in the bottom cover with a stainless steel connecting rod linkage.

- Single servo actuation eliminates complexity; servo placed centrally to minimise torque requirements.
- Link attachment point iteratively positioned near the free edge of the door to minimise required torque.
- Vertical retaining extensions added over the payload border to prevent payload displacement during flight manoeuvres.
- Payload drop triggered autonomously by the Python/ DroneKit algorithm running on the Raspberry Pi 4 flight computer.

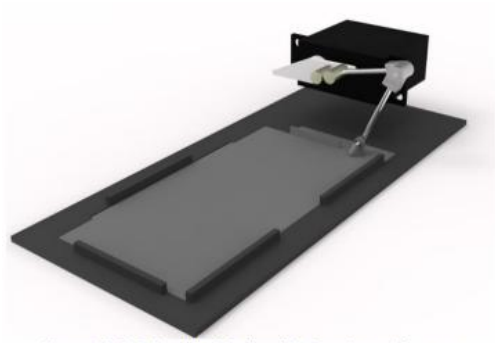


Figure 10 CAD Model of Payload Mechanism with servo

7. DESIGN OPTIMISATION & FINAL SPECIFICATIONS

7.1 Optimisation Changes from Preliminary to Final Design

After completing all structural FEA analysis, a systematic design review identified weight reduction opportunities across every major component without compromising safety margins. Total empty weight was reduced from ~1,820 g to 1,780 g.

Table 8 Design Changes

Component	Design Changes
Mid Structure	• Weight reduction on arm mounts based on analysis results. • Reducing the wall thickness from 5mm to 3mm without losing its strength. • Total weight reduced by 10%
+ Top and Base Plate	• The weight reduction on the plates were done based on the generation of stress. • As PMP met the requirements, HDPE was replaced by PMP.
Top Cover	• In the preliminary design, the entire top cover had to be removed to change the battery, thus a battery door that could fit on a single socket head screw was created to save time. • The weight was lowered by reducing the number of fasteners used.
Arms	• The diameter of the arms was decreased to 1mm since 2mm thick rods were overdesigned. • Total weight reduced by 25%
Landing gears	• According to the stress generation data, the square cross section was too safe, thus the I cross section was used, which produced the optimum results. • Total weight reduced

8. AUTONOMOUS SYSTEM ARCHITECTURE

The autonomous system integrates a Pixhawk Cube flight controller (inner loop: attitude) with a Raspberry Pi 4 flight computer (outer loop: navigation, object detection, payload decisions). Communication between them uses the MAVLink protocol over serial (921,600 baud). The entire system was simulated in ROS/Gazebo before physical implementation.

8.1 Avionics Stack

Flight controller	Pixhawk Cube — scored 77/80 in trade-off (vs. Pixhawk 2.4.8, AeroFC, APM 2.8)
Flight computer	Raspberry Pi 4 — scored 60/56 in trade-off for Processor, Weight, Size
Communication protocol	MAVLink via MAVProxy / DroneKit framework
Radio & video telemetry	Herelink integrated system — 2.4 GHz, up to 20 km range, HD video feed
GPS	Here GPS module (Pixhawk ecosystem)
FPV camera	Up to 2,000 m range
Object detection	YOLO v4-tiny on Raspberry Pi 4 — selected over YOLO v4, Fast R-CNN, Mobile SSD, haarCascades
Simulation environment	ROS/Gazebo with MAVros + DroneKit + QGroundControl

8.2 Object Detection Algorithm Selection

Five computer vision algorithms were evaluated on accuracy (mAP%), speed (FPS), and ease of implementation on the Raspberry Pi 4 hardware. YOLO v4-tiny was selected — optimal for single-class detection (target circle), scoring highest overall (19/25) while delivering real-time FPS performance on the Pi 4 without GPU acceleration.

A MATLAB Simulink state diagram modelled the complete autonomous pipeline: Takeoff → InAir → Search_Target → Target_detected → Payload_Drop_altitude → Drop → back_to_Ideal. This state machine was validated in simulation before flight testing.

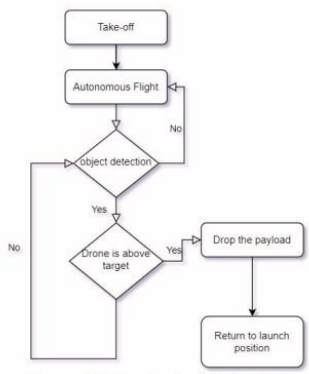


Figure 43 Flowchart of Payload drop mechanism of avionic components

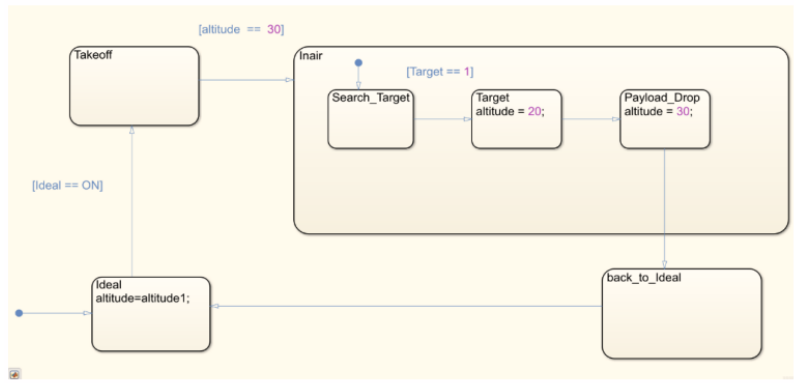


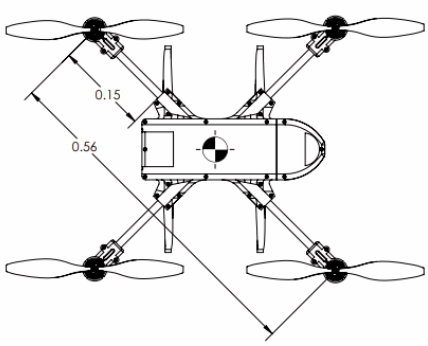
Figure 42 State Diagram simulation in Matlab of autonomous process

data

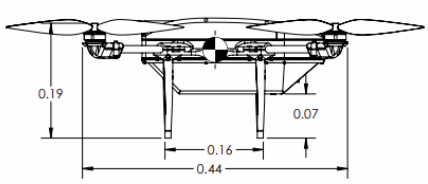
9. ENGINEERING DRAWING & Crash Prediction System

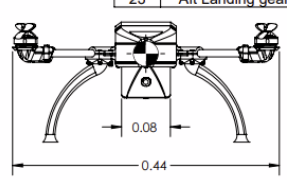
9.1 AI Crash Prediction System

- Trained a machine learning model using historical crash data, GPS telemetry, and IMU (flight controller position) data from manual flight sessions.
- System predicts a crash event with ≥ 60% confidence → instantly switches from manual to autonomous mode → drone auto-levels and lands safely.
- Addresses the primary failure mode in manual UAV operation competitions — pilot error during approach and landing phases.



Sl no.	Name	Weight (g)	Distance from datum (m)	Moment(g.m)
1	Top cover	88	0.19	16.92
2	Top plate	43	0.16	7.04
3	Mid plate	162	0.06	10.36
4	Front arms	36	1.15	41.69
5	Front Motor mounts	44	1.8	79.34
6	Front Motors	140	1.96	275.36
7	Base plate	20	0.22	4.53
8	GPS	40	1.32	52.86
9	ESC	10	0.62	6.23
10	Bottom cover	134	0.28	38.67
11	Raspberryy	50	0.95	47.59
12	FPV camer	6	1.27	7.65
13	Servo	65	0.86	56.23
14	Front Landing gear	58	0.78	41.47
15	Aft arms	36	-1.18	41.69
16	Aft motor mounts	44	-1.82	79.34
17	Aft Motors	140	-1.96	27.53
18	Battery	370	-0.37	139.19
19	Herelink	80	-0.95	76.52
20	Power module	25	-0.7	16.66
21	Payload	200	0	0
22	Pixhawk	75	0	0
23	Aft Landing gear	58	-0.78	45.67





Specifications	
Length	0.44m
Breadth	0.44m
Height	0.19m
Empty weight	1780g
Motor	Sunnysky V2806
Propeller	12 X 5 apc

	All dimensions are in meters	Team ID : AeroTHON 49
SCALE 1:5	aeroKLE Aero Design Team	Qty 1
SHEET A3	KLE Technological University Hubli - 31	SHEET 1/1

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